5.10 Converting Units Using Ratios

Lesson Introduction

Benchmark	MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths and conversions within the same measurement system.	Learning Target	• I can use ratios to convert units within the same measurement system.

	Building On	Working Toward
Benchmark Coherence	MA.5.M.1.1 Solve multi-step real-world problems that involve converting measurement units to equivalent measurements within a single system of measurement.	MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions. MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

Essential Questions	• How do you use ratios to convert units within the same measurement system? • How can you represent real-world scenarios using ratio conversions?
Lesson Narrative	In this lesson, students use unit rates to convert various measurements of customary and metric units within those systems. Students explore unit rate conversions for length, volume, mass, and time and apply unit rates to convert measurements. Students analyze errors in another student's work converting units of measure. Then, students practice converting units of measure using rates. Finally, students reflect on their level of confidence using ratios to convert units within the same measurement system.

Component Summary	5.10.1 Warm-Up (~5 min.)	Describe how furniture designers use math *(SE p. 219)*	5.10.4 Collaboration (10 min.)	Analyze errors in a student's work converting units of measure *(SE p. 222)*
	5.10.2 Guided Instruction (20 min.)	Use unit rate conversions for length, volume, mass, and time *(SE p. 220)*	5.10.5 Your Turn! (5 min.)	Practice converting units of measure using unit rates *(SE p. 223)*
	5.10.3 Guided Practice (5 min.)	Convert units of measure using unit rates *(SE p. 222)*	5.10.6 Wrap-Up (~5 min.)	Students reflect on their level of confidence using ratios to convert units within the same measurement system *(SE p. 224)*

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms</u>: • Customary units • Metric units <u>Additional Terms</u>: N/A	• (Optional) Note cards • (Optional) Chart paper • (Optional) Markers	• Warm-Up 5.10.1 contains a video to accompany the question prompts.

	Common Misconceptions	Things to Consider
Teacher Tips	• Students may make computational errors when converting units of measurement. • Students may accidentally use the reciprocal of the conversion factor.	• Students do not need to memorize conversion rates when engaging in problem-solving. Encourage students to use the tools they have (conversion tables provided as reference charts and in the Student Workbook) to solve problems efficiently.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share Consider structuring 5.10.3 Guided Practice and 5.10.5 Your Turn! with the Think Pair Share routine. First, have students solve the problem individually. Then, have students discuss their answers. Tell students to justify their responses and critique each other's reasoning. Discussing the problems will help students synthesize their understanding and build student confidence.	**MA.K12.MTR.2.1: Multiple Representations** In 5.10.2 Guided Instruction students use double number lines and equivalent ratio tables to convert units of measurement. Using these visual models will help students organize the quantities, determine the ratios, and accurately complete their calculations. Seeing multiple representations of the equivalent ratios will help students visualize the concepts and consolidate student understanding.
Differentiate Instruction	Consider alternative ways for students to engage with written descriptions or self-reflection like in 5.10.4 Collaboration and 5.10.6 Wrap-Up. Digital tools and apps like Flipgrid offer students differentiated approaches to communicating with others and capturing their personal reflections. 5.10.4 Collaboration: Have students verbally record their responses to Valerie using Flipgrid and submit as homework. 5.10.6 Wrap-Up: Have students describe what they learned with video, with more creative submissions earning bonus points. (encourage creativity in both examples)	
Lesson Notes:		

5.10.1 Warm-Up: Furniture Designer

Component Overview: *Display the "Career Profile – Furniture Designer" video for the entire class. After students watch the video, prompt the students to read and answer the questions in the Warm-Up independently. Discuss how the furniture designer used math in the video with the class.*

5.10.1 Warm-Up: Furniture Designer

1. Jared Rusten is a woodworker who creates modern furniture in San Francisco.

   a. Jared usually starts designing furniture by sketching his ideas before he makes a scaled drawing using measurements to build the actual piece of furniture. What do you think a scale drawing is?
   Answers will vary.

   A scaled drawing shows a real object that has accurate sizes that are reduced or enlarged by a scale or specific amount.

   b. Jared said, "I never thought that I'd be using math as much as I do now." What are some of the ways he uses math as a furniture designer and maker?
   Answers will vary.

   Scale drawings allow us to visualize larger objects on paper. As a furniture designer and maker, you draft or sketch your designs on paper before creating the pieces. You are also able to scale your designs before they are in actual spaces. He multiplies to enlarge, divides to reduce, and uses area to ensure that the furniture fits in a space.

Turn and Talk
Engage students in a Turn and Talk, after watching the video, to discuss question 1. Position students to consider the different ways in which Jared used math to create furniture. This will increase investment and engagement with the lesson.

5.10.2 Guided Instruction: Converting Units of Measure Using Ratio Relationships

Component Overview: *Students name different customary units of measurement for length, weight, volume, and time and find their unit conversions. Guide students through this component to explore how to convert units of measurement using the methods learned throughout this unit such as double number lines and tables.*

5.10.2 Guided Instruction: Converting Units of Measure Using Ratio Relationships

Many different units are used to measure lengths, weight, capacity (volume), and time.

1. What are some units of measure that are used to take each type of measurement?
   Answers will vary, but could include:

Length	Weight	Capacity	Time
Centimeter	*Ounce*	*Liter*	*Second*
Foot	*Pound*	cm^3	*Minute*
Mile	*Kilogram*	*Quart*	*Hour*
Inch	*Gram*	*Pint*	*Day*
Meter	*Ton*	*Gallon*	*Week*
Yard		*Cup*	*Year*

The United States uses the customary system of measurement. The table shows the unit conversions for units within this system.

2. Label each row using the words **Length, Weight,** and **Capacity** to indicate what each set of units measures.

Type of Measurement	Unit Conversions in the Customary System
Length	$1\ foot\ (ft) = 12\ inches\ (in.)$ $1\ yard\ (yd) = 3\ feet\ (ft)$ $1\ mile\ (mi) = 5{,}280\ feet\ (ft)$ $1\ mile\ (mi) = 1{,}760\ yards\ (yd)$
Capacity	$1\ cup\ (c) = 8\ fluid\ ounces\ (fl\ oz)$ $1\ pint\ (pt) = 2\ cups\ (c)$ $1\ quart\ (qt) = 2\ pints\ (pt)$ $1\ gallon\ (gal) = 4\ quarts\ (qt)$
Weight	$1\ pound\ (lb) = 16\ ounces\ (oz)$ $1\ ton = 2{,}000\ pounds\ (lb)$

Consider activating student interest by prompting one of the following:

- *What types of units of measure do we think Jared (from the 5.10.1 Warm-Up) might use on a daily basis?*
- *Brainstorm some units of measure that your family members might use on a daily basis in their jobs.*
- *What is your dream job? What units of measure are used most often in that environment?*

5.10.2 Guided Instruction: Converting Units of Measure Using Ratio Relationships (continued)

Each conversion rate represents a ratio.
Double number line diagrams can be used to convert units.

3. Use the double number line diagram shown to find how many feet (ft) are in 4 yards (yd).

There are 12 feet in 4 yards.

Tables can also be used to convert units of measure.

4. The average weight of an African forest elephant is 6,000 pounds. Use the unit conversion and the table to determine how many tons the average African forest elephant weighs.

Pounds	Tons
2,000	1
6,000	3

Notice that each conversion rate is a unit rate.
Unit rates can be used as factors to convert from one unit to another.

For example, the factor $\frac{4\ quarts}{1\ gallon}$ is equivalent to 1 because the numerator and denominator are equivalent values. Multiplying by factors equivalent to 1 will result in equivalent values.

Think-Pair-Share
Consider implementing a Think-Pair-Share before students use the representations to convert the units.
Pose the following questions to students to activate prior knowledge and allow students to verbalize their thinking, which will engage auditory learners.
How can you use the double number line to determine how many feet are in 4 yards?
How can you use the equivalent ratio table to determine how many tons the average African forest elephant weighs?

5.10.2 Guided Instruction: Converting Units of Measure Using Ratio Relationships (continued)

Kaluba determined how many quarts are in 2 gallons of milk using this strategy. His work is shown.

$$\frac{2\,gal}{1} \times \frac{4\,qt}{1\,gal} = 8\,qt$$

Kaluba explained, "In my calculation, the unit 'gallons' is in the numerator and the denominator. This means I can eliminate this unit because it becomes 1, similar to when equivalent values are in the numerator and denominator. The result of the conversion is then the number of quarts."

5. Complete the calculation to find how many fluid ounces (fl oz) are in 3 pints (pt).

$$\frac{3\,pt}{1} \times \frac{2\,cups}{1\,pt} \times \frac{8\,fl\,oz}{1\,cup} = 48\,fl\,oz$$

The units of measure used in the United States are ***customary units***.

The units of measure used in most of the world are ***metric units***. Like the decimal system, the metric system uses the base 10.

The tables below show additional unit conversion rates for other units of measure.

Type of Measurement	Unit Conversions in the Metric System
Length	1 meter (m) $=$ 100 centimeters (cm) $1\,meter\,(m) = 1000\,millimeters\,(mm)$ $1\,kilometer\,(km) = 1{,}000\,meters\,(m)$
Capacity	$1\,liter\,(L) = 1000\,milliliters\,(mL)$
Weight or Mass	$1\,gram\,(g) = 1000\,milligrams\,(mg)$ $1\,kilogram\,(kg) = 1{,}000\,grams\,(g)$
Time Conversions	
$1\,minute\,(min) = 60\,seconds\,(s)$ $1\,hour\,(h) = 60\,minutes\,(min)$ $1\,day = 24\,hours\,(h)$ $1\,year = 365\,days$ $1\,year = 52\,weeks$ $1\,week = 7\,days$	

Challenge students who may finish early to convert units using the metric system or units of time. Consider asking students to determine the number of minutes in a week or a year (like in the musical Rent – 525,600). This will provide additional practice with more difficult conversion rates.

5.10.3 Guided Practice: Converting Units of Measure Using Unit Rates

Component Overview: *Students work with a partner to convert units of measurement using unit rates.*

5.10.3 Guided Practice: Converting Units of Measure Using Unit Rates

1. Use the conversion rates to find each value. Show your work.

   a. 2.62 km $= 2620$ m

   b. 9 in $= \frac{1}{4}$ yd

   c. 1 week $= 168$ hours

2. Adrian weighed 7 pounds (lb) and 4 ounces (oz) when he was born. How many ounces did Adrian weigh when he was born?
   116 *oz*

3. Ibuprofen is a medicine sometimes used for pain relief, such as for a headache. For adults, it is not safe to take more than 3200 milligrams (mg) of ibuprofen each day. How many grams of ibuprofen is safe for adults to take in one day?
   3.2 *grams*

Think-Pair-Share for MA.K12.MTR.1.1 Practice
Consider structuring this component with the Think-Pair-Share routine.

- *First, have students solve the problem individually, emphasizing their process rather than their product.*
- *Then, have students discuss their answers. Tell students to justify their responses and critique each other's reasoning.*
- *Discussing the problems will help students synthesize their understanding and build student confidence.*

5.10.4 Collaboration: Error Analysis

Component Overview: *Students work with a partner or small group to analyze errors in a student's work converting units of measure.*

5.10.4 Collaboration: Error Analysis

1. Valerie makes sure she drinks 8 cups (c) of water each day. She wondered how many gallons of water this is, so she used conversion rates to find out. Her work is shown.

$$\frac{80\ c}{1} \times \frac{2}{1} \times \frac{1}{2} \times \frac{1}{4} = \frac{160}{8} = 20\ gal$$

Valerie knew her answer didn't make sense because there is no way she drinks 20 gallons of water each day. She is not sure where she went wrong.

a. Discuss Valerie's work with your partner. Then, work together to find the correct number of gallons she drinks each day.

Valerie's first mistake was inputting 80 cups instead of her daily intake of 8 cups. Also, Valerie's first conversion factor, $\frac{2}{1}$, should be $\frac{1}{2}$ because there are 2 cups in 1 pint.

The correct answer is:

$$\frac{8\ \cancel{c}}{1} \times \frac{1\ \cancel{pt}}{2\ \cancel{c}} \times \frac{1\ \cancel{qt}}{2\ \cancel{pt}} \times \frac{1\ gal}{4\ \cancel{qt}} = \frac{8\ gal}{16} = 0.5\ gal$$

If students have not mastered the multiplication and division facts, let them use a calculator to determine the ratios rates, and percentages. The calculator will allow students to focus on mastering the learning goals of the lesson.

Circulate as students work with their partners to make informal observations and listen as students engage in discourse. Students should be using math vocabulary to find and explain the errors in Valerie's work, such as conversion, unit rate, and ratio.

5.10.4 Collaboration: Error Analysis (continued)

b. Help Valerie understand and learn from her mistakes. Write a brief note to Valerie explaining what her mistakes were and how she can learn from them. Include strategies that may be helpful.

Hello Valerie. I see that you do know your conversions! The one mistake you made in this problem is in the first conversion factor, $\frac{2}{1}$, which should be $\frac{1}{2}$ because there are 2 c in 1 pt. By writing $\frac{2}{1}$, you will not be able to eliminate the cups unit. One strategy that may be helpful is to include the units when writing the conversion factors. This will help you see that you are eliminating the correct units and using the correct conversion factors.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Students may struggle to explain Valerie's mistakes in writing or struggle to explain a strategy to fix them. Ask probing questions to guide student thinking or convert these questions into sentence stems for students to guide their responses.

- *What did Valerie do wrong? How did you fix that?*
- *What if you complete the question on your own and compare your work to Valerie's? Find the differences.*
- *How can you use units to help convert measurements correctly?*

c. Read your partner's note to Valerie and offer suggestions on ways to improve their note.
Answers will vary.

5.10.5 Your Turn!

Component Overview: *Students work independently to practice converting units of measure using unit rates.*

5.10.5 Your Turn!

1. Use the conversion rates to find each value. Show your work.

   a. $11{,}200$ milliliters $= 11.2$ liters

   b. 3.5 gallons $= 56$ cups

   c. 2 days $= 2{,}880$ minutes

2. Malik plays football for Jones High School in Orlando, Florida. Last Friday, he ran 94 yards for a touchdown on a kickoff return. How many feet did Malik run on the kickoff return?

$$\frac{94\ yd}{1} \times \frac{3\ ft}{1\ yd} = \frac{282\ ft}{1} = 282\ ft$$

3. About 727 pints of milk are distributed each day from the cafeteria at Fort King Middle School in Marion County, Florida. How many gallons of milk are distributed at Fort King Middle School each day?

$$\frac{727\ pt}{1} \times \frac{1\ qt}{2\ pt} \times \frac{1\ gal}{4\ qt} = \frac{727\ gal}{8} = 90.875\ gal$$

To encourage mathematical reasoning as students complete this component, ask questions like:

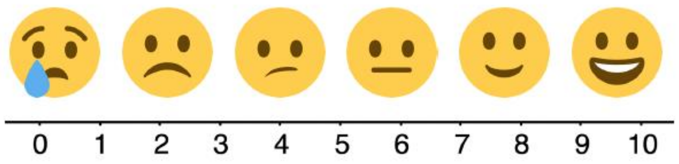
- *How can you determine the reasonableness of your answer using unit rates?*
- *How can you estimate what your answer should be before converting? Why would this be helpful?*
- *Will Malik have run more in feet or yards? How can you use this to assess the reasonableness of your response?*

5.10.6 Wrap-Up: Reflection

Component Overview: *Consider using all or parts of this exercise as a formative assessment to gather data on student progress from this lesson. This Wrap-Up assesses students' level of confidence with using ratios to convert units within the same measurement system.*

5.10.6 Wrap-Up: Reflection

1. Use the emoji scale to indicate your level of confidence with using ratios to convert units within the same measurement system.



Answers will vary.

2. Explain why you indicated the confidence level above.
Answers will vary.

Encourage students to get creative and represent their learning or progression from the lesson using video app platforms, drawings, or other methods and submit as homework.